

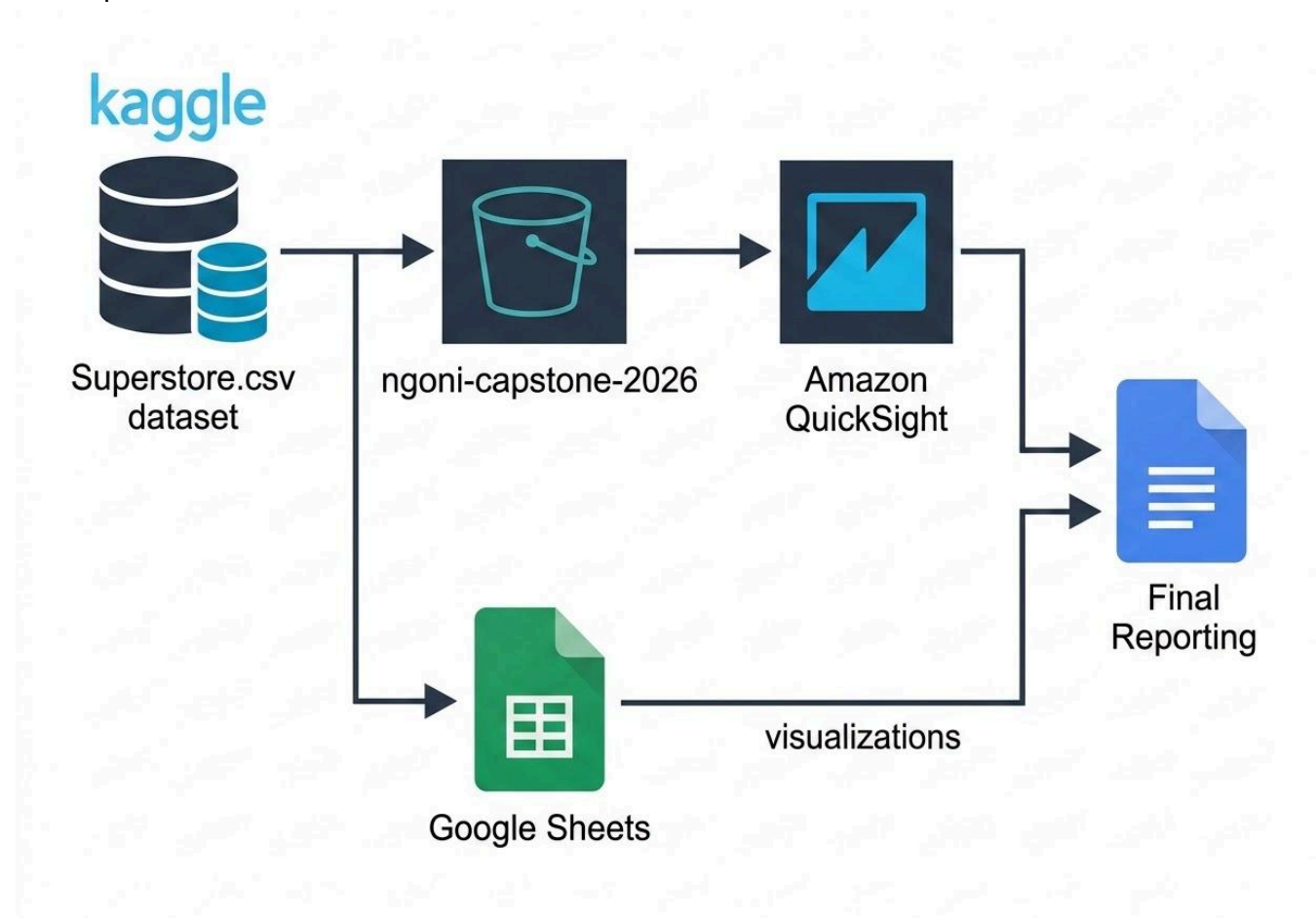
Name : Ngonidzashe Roselyne Kazonda

Email Address : kazondarosy@gmail.com

Cloud Foundation Virtual Bootcamp - Final Capstone Project Data Analyst

Technical Architecture

The technical pipeline for this Capstone Project spans both AWS Cloud Infrastructure and Google Workspace to support secure storage, advanced analytics, and professional dashboard presentation. The diagram below illustrates the end-to-end data flow from source to final presentation:



Data Preparation

The analysis began by downloading the Superstore sales dataset from Kaggle. The downloaded files were then extracted locally to prepare the dataset for processing and cleaning in Google Sheets.

Upon uploading the dataset into Google Sheets, an initial review was conducted to identify data cleaning requirements and transition the dataset from raw to clean data:

- **Row ID Filtering:** Filters were applied to the Row ID column to remove blank spaces.
- **Currency Formatting:** The currency values were converted from British pounds to US dollars (USD).
- **Discount Formatting:** The discount column format was updated to percentage values.

Analysis of the sales performance

1. Consistent Annual Growth in Sales and Profitability

- **Strong Upward Trend:** Total sales have steadily increased year-over-year, growing from \$484,247.52 in 2011 to \$733,947.13 in 2014.
- **Profit Expansion:** Annual profits followed a similar positive trajectory, expanding from \$49,543.99 in 2011 to \$93,507.51 in 2014, demonstrating healthy operational scaling.
- **Core Drivers:** As shown in the 'Sales and Profit by Category' chart, Technology and Office Supplies are the primary drivers of this profitability, contributing \$145,454.95 and \$122,490.80 in total profits respectively.

2. Profitability Erosion in Furniture Due to Heavy Discounting

- **High Volume, Low Return:** Despite generating substantial sales (\$741,999.76), the Furniture category yielded only \$18,451.29 in total profit, resulting in a meager 2.49% profit margin.
- **Unprofitable Sub-Categories:** The 'Total Profit by Sub-Category' chart highlights that Tables (-\$17,725.47) and Bookcases (-\$3,472.55) are significantly in the red, dragging down the entire category's performance.
- **Discount Correlation:** This poor performance directly correlates with aggressive promotional pricing. Tables and Bookcases carry the highest average discounts at 26.13% and 21.11% respectively, which severely erodes product margins.

3. Regional Performance is Heavily Impacted by Discounting Strategies

- **West and East Lead:** The West and East regions are the top contributors, with the West region generating \$725,457.86 in sales and \$108,418.44 in profit, and the East region bringing in \$678,781.31 in sales and \$91,522.79 in profit.
- **Central Region Margin Squeeze:** The Central region generated higher sales than the South region (\$501,239.93 vs. \$391,721.93) but brought in lower absolute profits (\$39,706.38 vs. \$46,749.43).
- **The Discount Impact:** The 'Profit Margin vs. Average Discount by Region' chart reveals that the Central region has the highest average discount rate (24.04%), which directly suppresses its profit margin to a low of 7.92%. Conversely, the West

region maintains the lowest average discount (10.93%) and achieves the highest profit margin (14.94%).

Identification of top-performing products

The top-performing products in the dataset were evaluated in two ways : **Total Profit and Total Sales**

The following products were the most valuable because they generated a lot of revenue while maintaining excellent profit margins

- Canon imageClass 2200 Advanced Copier : It was the single best-performing product in the entire catalog .
It generated \$61,599.98 in total sales and brought in \$25,199.94 in profits.
- Fellowes PB500 Electric Punch Plastic Comb Binding Machine : It produced \$27,753.03 in sales and \$7,753.03 in profits
- Hewlett Packard LaserJet 3310 Copier : It produced \$18,839.69 in sales and \$6,983.89 in profit.

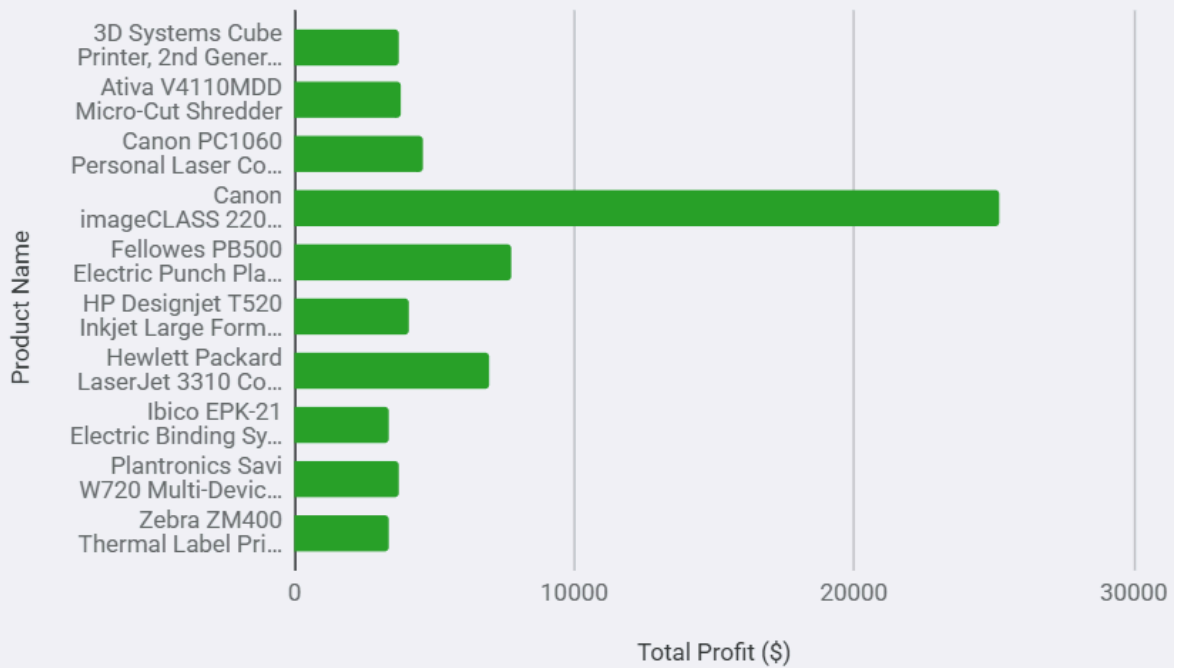
Some of the products actually lost money due to heavy discounting or high costs :

- Cisco TelePresence System EX90 videoconferencing Unit : It produced \$22,638.50 in sales but resulted in a net loss of -\$1,811.08
- GBC DocuBind P400 Electric Binding System : It produced \$17,965.07 in sales but lost -\$1,878.17 in profit
- HON 5400 Series Task Chairs for Big and Tall : \$21,870.57 in sales but yielded \$0.00 in profit.

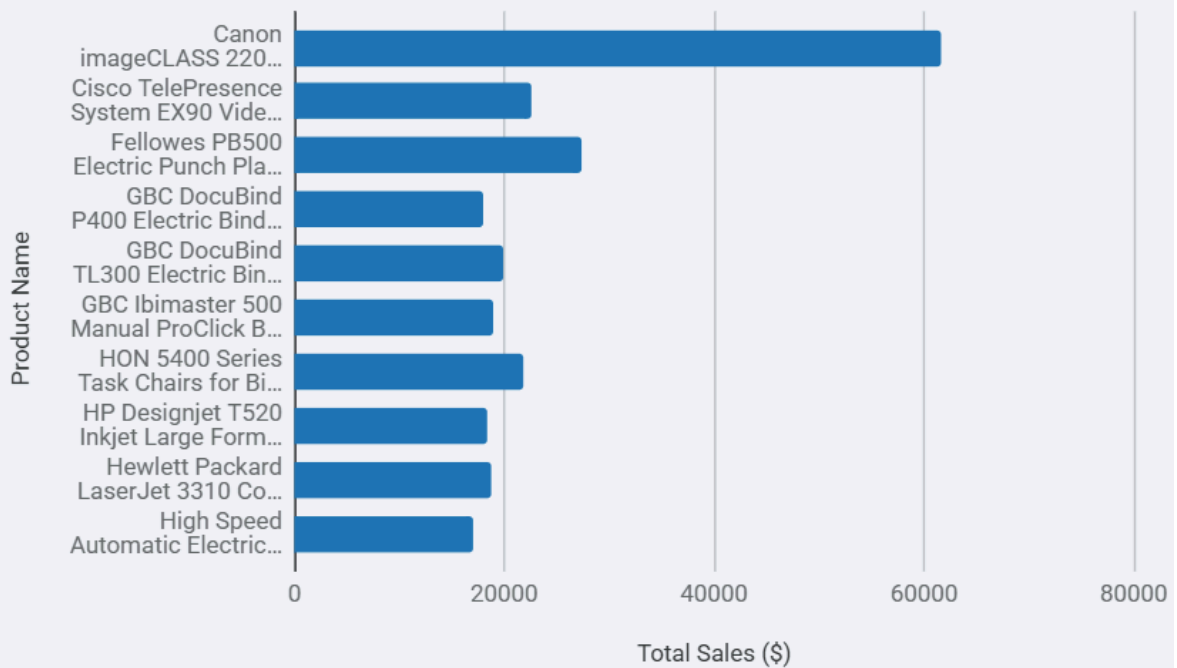
These products didn't have the highest sales volumes , but they were efficient at converting their revenue into profit :

- Canon PC1060 Personal Laser Copier : It produced \$11,619.83 in sales and \$4,570.94 in profit (39% profit margin)
- Ativa V4110MDD Micro-Cut Shredder : It produced \$7,699.89 in sales and yielded \$3,772.95 in profit (49% profit margin)

Top 10 Most Profitable Products



Top 10 Products by Total Sales



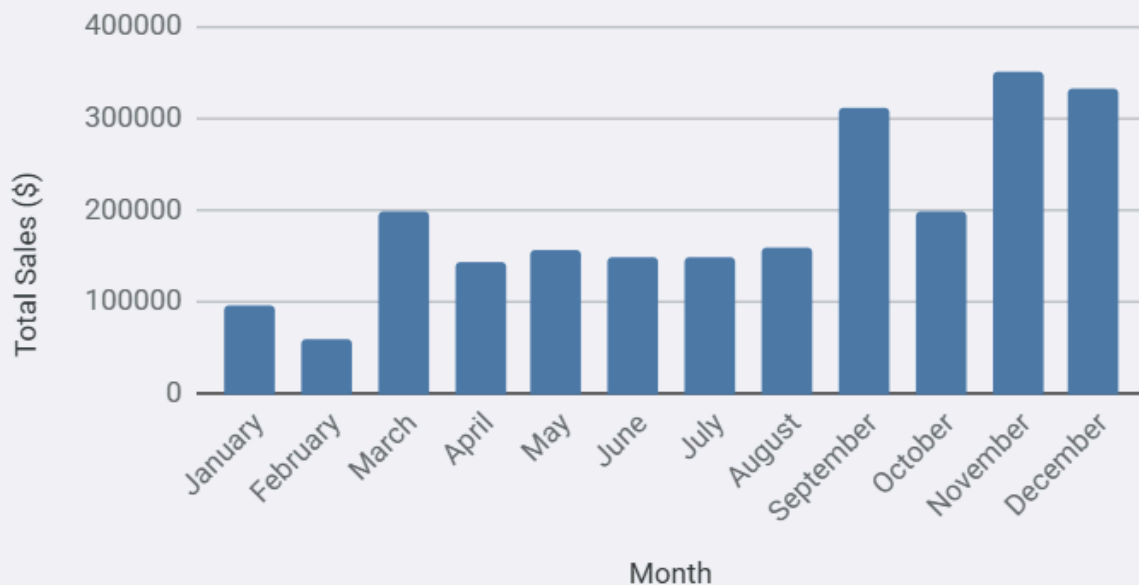
Identification of trends

1. Steady Year-over-Year Business Growth
 - The business was expanding consistently, showing strong upward momentum in both revenue and profits
 - Rising Sales : Total annual sales grew from \$484,248 in 2011 to \$733,947 in 2014, representing a significant expansion in market reach.
 - Increasing Profits : Annual profits nearly doubled from \$49,544 in 2011 to \$93,508 in 2014
 - Higher Volume : The total quantity of items sold increased every single year, moving from 7,581 items in 2011 to 12,503 items in 2014



2. Strong End-of-Year Seasonality : The business experienced a high predictable seasonal cycle, with a massive surge in activity during the fall and winter months

Monthly Sales Seasonality (All Years Combined)



3. The Furniture Profitability Challenge : Some product categories were highly efficient at turning sales into profit , others were struggling
 - High Sales, Low Return - The Furniture category produced \$741,999 in revenue but yielded a small fraction in profit \$ 18,451
 - The Profit Champions : Technology produced \$836,154 in sales but produced \$145,455 in profit. Office Supplies produced \$719,047 in sales and produced \$122,491 in profit
 - The Hidden Culprits : Tables (losing \$17,725) and Bookcases (losing \$3,473) were draining money from the business due to steep discounting or high shipping costs

Business Recommendations

Implement targeted discount policies :

- Reviewing Furniture Pricing : The furniture category was a profitability drain due to aggressive discount
 - Move away from high-volume, low-margin strategies for these products
 - Implement a more disciplined discount policy to protect margins
- Focus on High-Efficiency Products : Prioritize marketing and inventory resources for products that demonstrate high revenue-to-profit conversion such as the Canon PC1060 Personal Laser Copier (39% margin) and Ativa V4110MDD Micro-Cut Shredder (49% margin)

- Phase out or reposition underperformers : Evaluate the future of products currently resulting in net losses
- Capitalize on Seasonal Trends : Prepare for Seasonal Surges - The data reveals a predictable and significant surge in activity during the fall and winter months.
 - Ensuring inventory levels, warehouse staffing, and customer support resources are fully scaled to meet this peak demand to maximize revenue capture without service degradation
- Build on Technology and Office Supplies Success : These categories are primary drivers of profitability

Dashboard

Total Sales

\$2,297,200.86

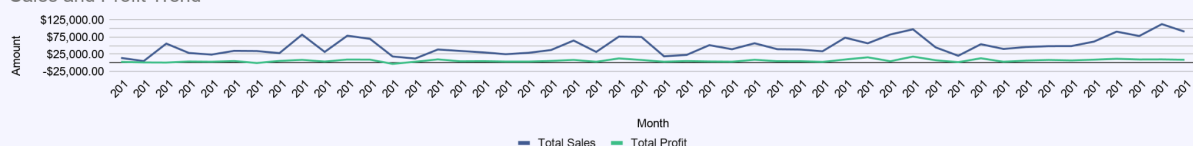
Total Profit

\$286,397.02

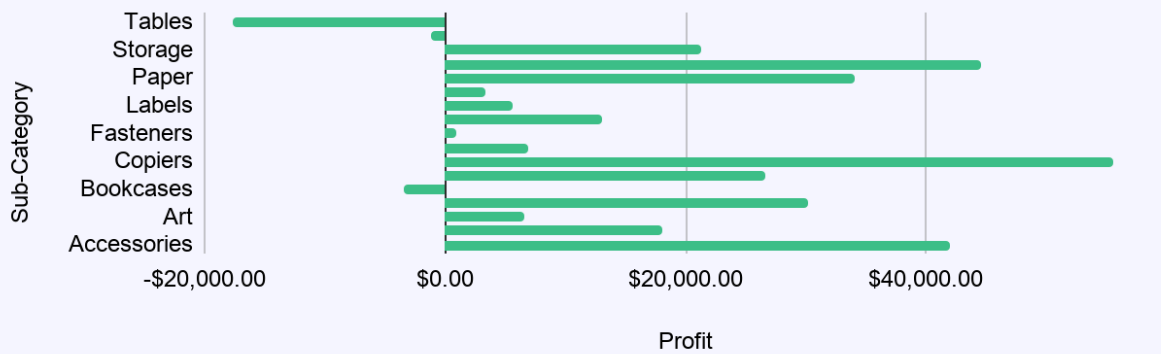
Profit Margin

12.47%

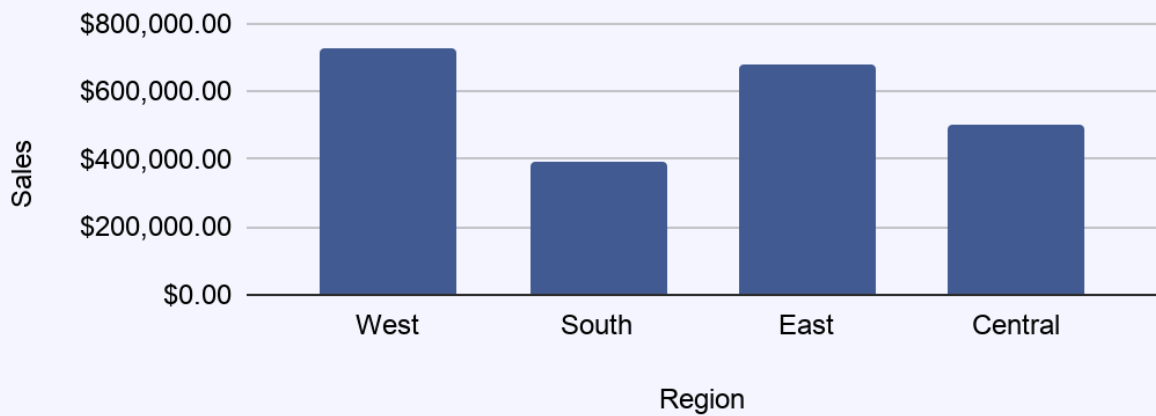
Sales and Profit Trend



Profit by Sub-Category



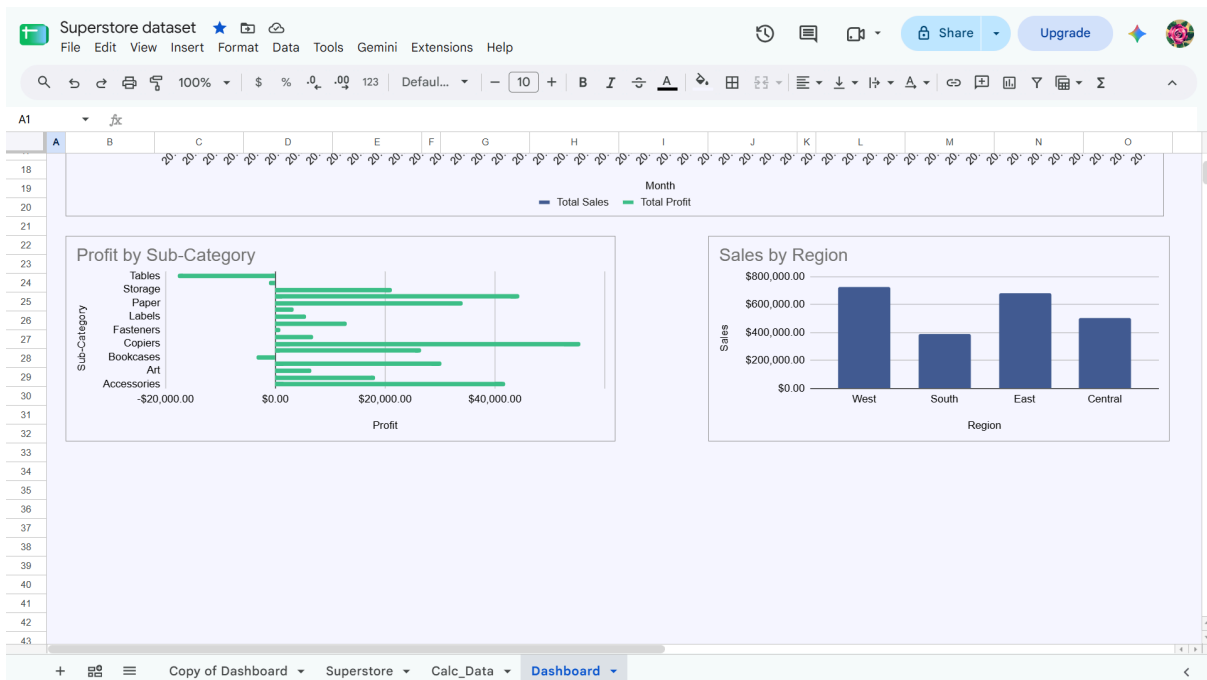
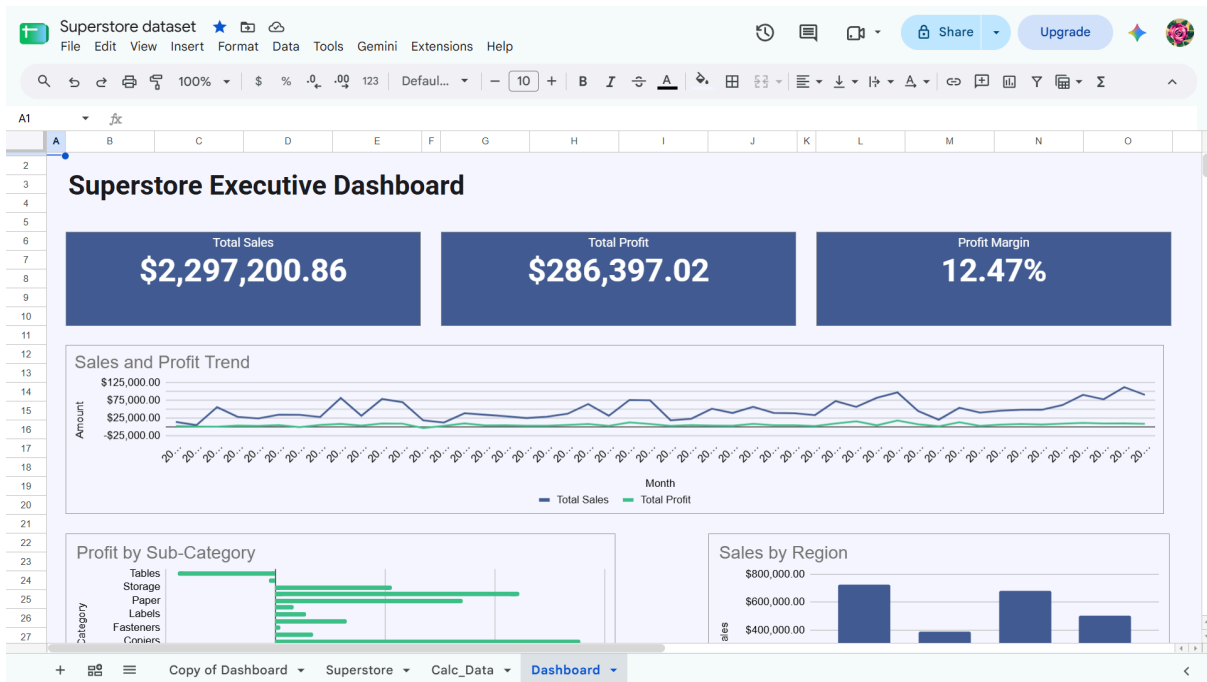
Sales by Region



Profit Margin

12.47%

Screenshots of my dashboard in Google Sheets



Screenshots of the work I did on AWS

Amazon S3

 >

Buckets

 >

ngoni-capstone-2026

✔ Successfully created bucket "ngoni-capstone-2026"

✕

ngoni-capstone-2026

Info

<

Objects

 Metadata Properties Permissions Metrics Management File sys >

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

Privacy

Terms

Cookie preferences

© 2026, Amazon Web Services, Inc. or its affiliates.

Amazon S3

✔ Upload succeeded

✕

For more information, see the **Files and folders** table.

Files and folders (1 total, 2.2 MB)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
Superstore.csv	-	text/csv	2.2 MB	✔ Succeeded	-

CloudShell

Agent Toolkit for AWS

Feedback

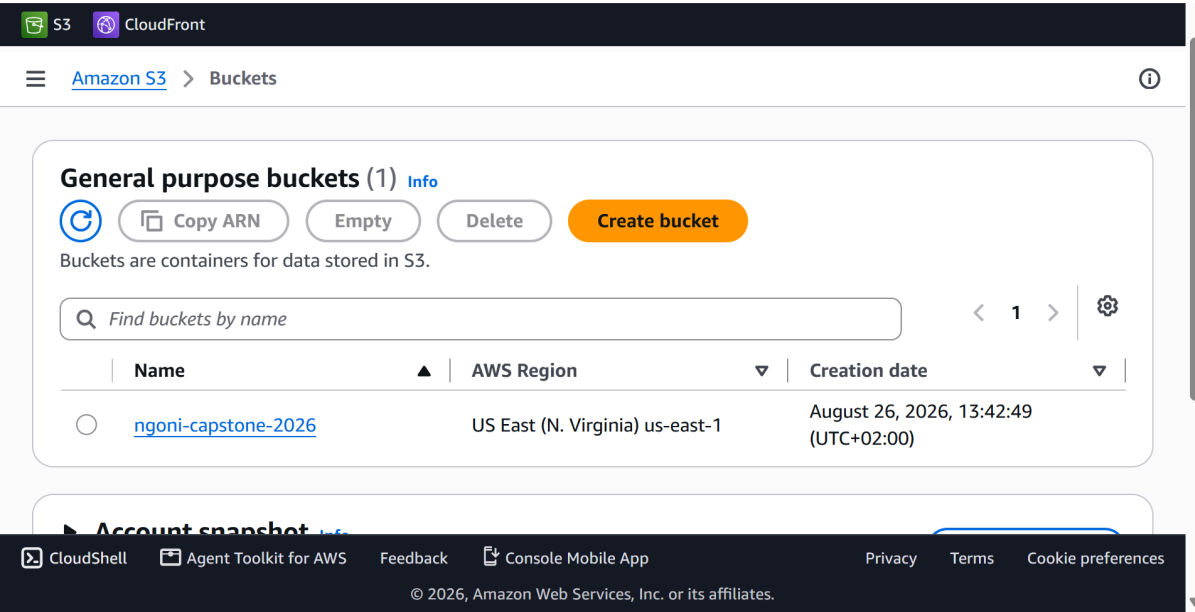
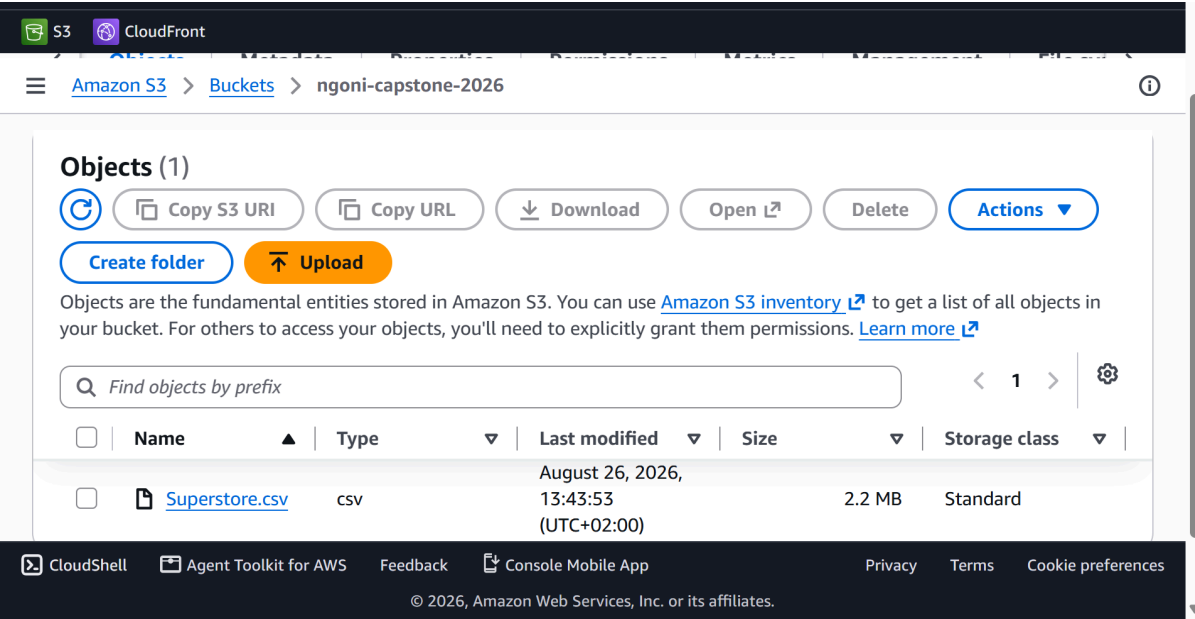
Console Mobile App

Privacy

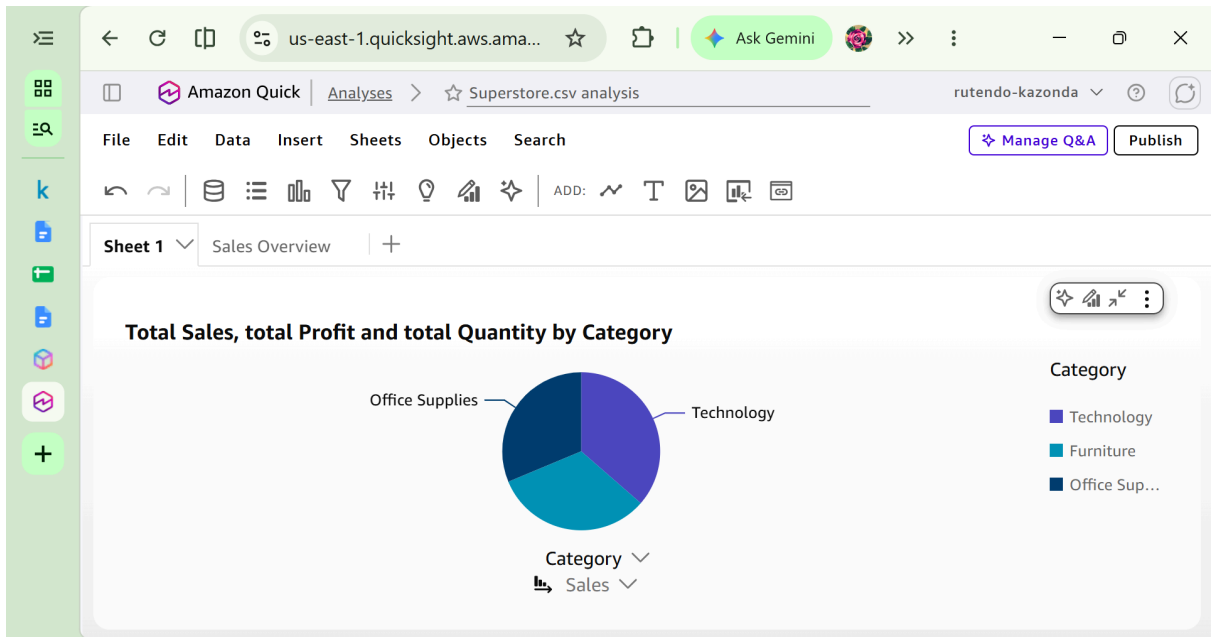
Terms

Cookie preferences

© 2026, Amazon Web Services, Inc. or its affiliates.



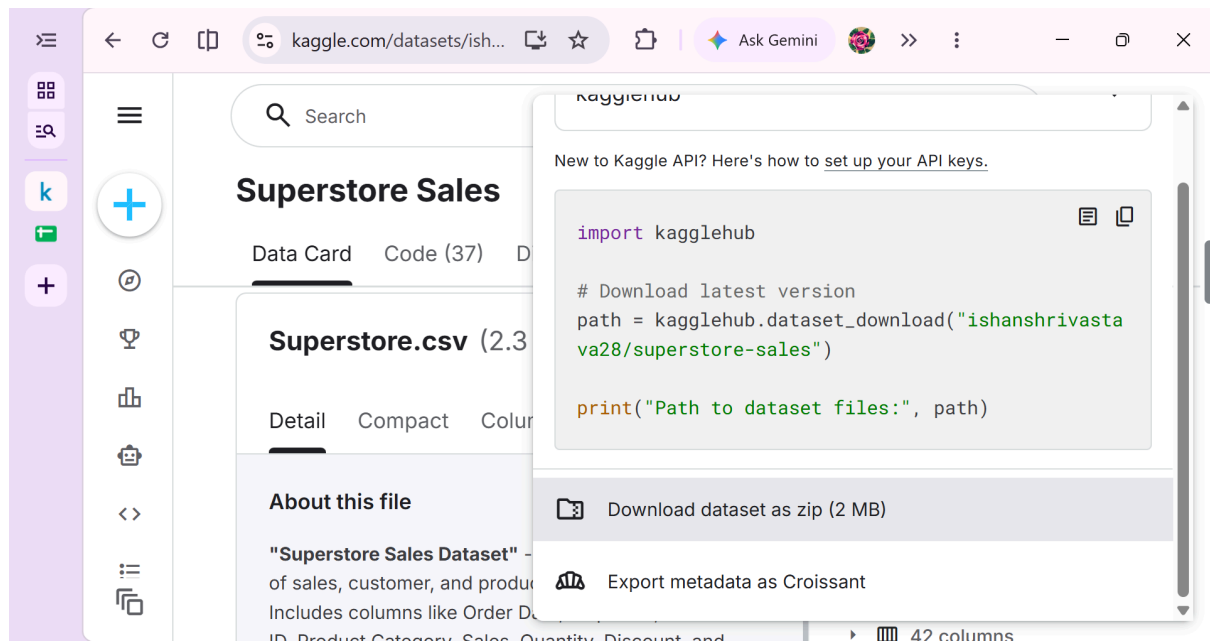
The visualizations I made with Quick Sight



Screenshots from Kaggle

The screenshot shows the Kaggle website interface for the 'Superstore Sales' dataset. The browser address bar displays 'kaggle.com/datasets/ish...'. The page title is 'Superstore Sales'. The dataset has 215 rows and 42 columns. The 'Data Card' tab is selected, showing a table of the dataset. The 'Data Explorer' tab is also visible, showing a tree view of the dataset files: Superstore.csv, Superstore.xlsx, and Orders. The 'Summary' section shows that the dataset contains 2 files and 42 columns. The 'About this file' section provides a description of the dataset: 'Superstore Sales Dataset' - Comprehensive collection of sales, customer, and product data for analysis. Includes columns like Order Date, Ship Date, Customer ID, Product Category, Sales, Quantity, Discount, and...

File	Size
Superstore.csv	2.3 MB
Superstore.xlsx	3.63 MB
Orders	



Let's Connect

Github - <https://github.com/Rosy70bit>

Linkedin - www.linkedin.com/in/ngonidzashe-kazonda-6a270038a

Email Address - kazondarosy@gmail.com